Kudeki

Exam 1

Thu, June 28, 2018 — 11 AM - 11:50 AM

Name:	Solution
Section:	

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the above boxes.

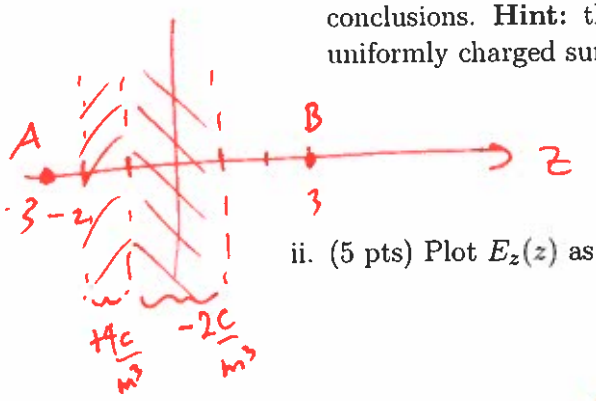
This is a closed book exam and calculators are not allowed. You are allowed one formula sheet of 8.5 by 11 inch dimensions — both sides of the sheet may be used. Please show all your work and make sure to include your reasoning for each answer. All answers should include units wherever appropriate.

Problem 1 (25 points)	
Problem 2 (25 points)	
Problem 3 (25 points)	
TOTAL (75 points)	

1. Three parts of this question are independent:

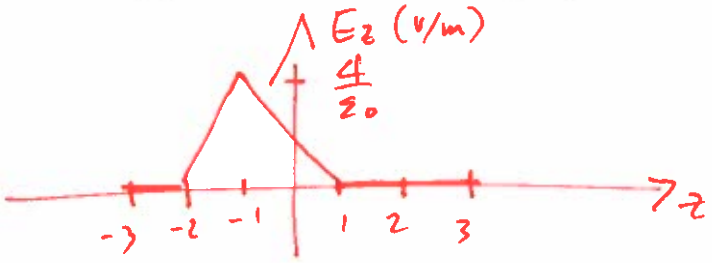
   a) We have two infinite slabs of space charge of finite widths in the z-direction. Slab 1 occupies the region $-2 < z < -1$ m and has a charge density of $\rho_1 = 4$ C/m^3 while Slab 2 occupies the region $-1 < z < 1$ m and has a charge density of $\rho_2 = -2$ C/m^3.

      i. (5 pts) Determine the electric field vectors $\mathbf{E}_A$ and $\mathbf{E}_B$ at locations $A = (x, y, z) = (0, 0, -3)$ m and $B = (x, y, z) = (0, 0, 3)$ m. Show your work and explain briefly how you reach your conclusions. **Hint:** the field outside a uniformly charged slab is the same as the field of a uniformly charged surface having the same amount of C/m^2!



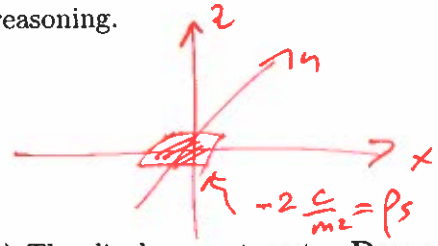
$\bar{E}_A = \bar{E}_B = 0$ because $|4\times1| = |-2\times2|$

      ii. (5 pts) Plot $E_z(z)$ as a function of z for $-3 < z < 3$ m and justify your plot?



   b) A square patch of 1 m × 1 m dimensions centered about the origin on the $z = 0$ plane supports a surface charge density of $\rho_s = -2$ C/m^2. Provide reasonable approximations for:

      i. (5 pts) The displacement vector $\mathbf{D} = \epsilon_o\mathbf{E}$ on the z-axis 1 cm above the patch — explain your reasoning.



$\bar{D} = +\hat{z}\,\frac{\rho_s}{2} = -\hat{z}\,\frac{-2\ \mathrm{C/m^2}}{2} = -\hat{z}\,\frac{\mathrm{C}}{\mathrm{m^2}}$

very close to the patch use infinite sheet formula

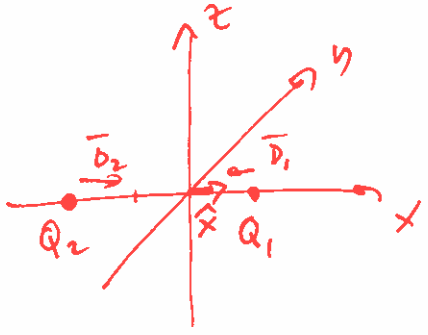
      ii. (5 pts) The displacement vector $\mathbf{D} = \epsilon_o\mathbf{E}$ on the z-axis 1 km above the patch — explain your reasoning.

$\bar{D} = +\hat{z}\,\frac{Q_s}{4\pi r^2} = -\hat{z}\,\frac{2}{4\pi\,10^6} = \frac{-\hat{z}}{2\pi\times10^6}\,\frac{\mathrm{C}}{\mathrm{m^2}}$

$Q_s = (-2\ \mathrm{C/m^2})(1\ \mathrm{m^2})$

$r^2 = (10^3)^2$

   c) (5 pts) Two point charges, $Q_1 = 2$ C and $Q_2 = -2$ C, are located at $(x, y, z) =$(1, 0, 0) m and (-2, 0, 0) m, respectively. What is the displacement flux $\int_S \mathbf{D}\cdot\hat{x}dydz$ through the entire $x = 0$ surface S? Explain.



$\int_S \bar{D}\cdot\hat{x}\,dydz = \frac{Q_2}{2} - \frac{Q_1}{2} = \frac{-2}{2} - \frac{2}{2}$

$= -1 - 1 = -2$ C

2. Three parts of the following question are independent:

a) Consider the electric field $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2$, where $\mathbf{E}_1 = \hat{x}\sin(\pi x/2)$ V/m and $\mathbf{E}_2 = \hat{x}\sin(\pi y/2)$ V/m.

i. (4 pts) Which one of $\mathbf{E}_1$ and $\mathbf{E}_2$ satisfies the electrostatic field conditions? Explain.

$\nabla \times \bar{E}_1 = 0$, $\nabla \times \bar{E}_2 \neq 0 \Rightarrow \therefore \bar{E}_1$ satisfies the electrostatic condition

ii. (4 pts) What is the charge density $\rho(x, y, z)$ in the region?

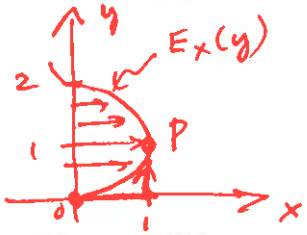
$\frac{\rho}{\varepsilon_0} = \nabla \cdot \bar{E} = \nabla \cdot \bar{E}_1 \Rightarrow \rho = \varepsilon_0 \nabla \cdot \bar{E}_1 = \varepsilon_0 \cos\left(\frac{\pi x}{2}\right)\frac{\pi}{2}\ \frac{C}{m^2}$ ($\nabla \cdot E_1$)

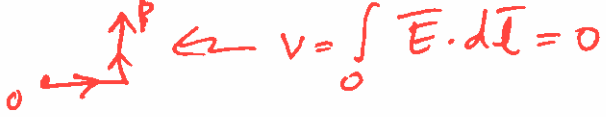
iii. (4 pts) What is the maximum value of "flux integral per unit volume" of the **E**-field in the region? Explain your answer carefully.

$\rightarrow |\nabla \cdot \bar{E}|_{max} = \left|\cos\left(\frac{\pi y}{2}\right)\frac{\pi}{2}\right|_{max} = \frac{\pi}{2}\ \frac{V}{m^2}$

b) Consider the field $\mathbf{E} = \hat{x}\sin(\pi y/2)$ V/m and the path dependent line integral $V = \int_O^P \mathbf{E} \cdot d\mathbf{l}$, where $O = (0, 0, 0)$ and $P = (1, 1, 0)$ m.

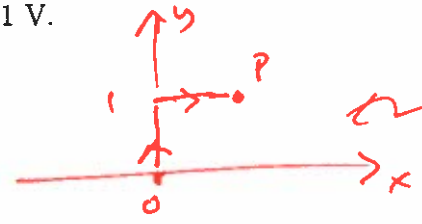
i. (3 pts) Make the sketch of an integration path from O to P on the xy-plane for which the line integral $V = 0$. Explain your reasoning.





$\leftarrow V = \int_O^P \bar{E} \cdot d\bar{l} = 0$

ii. (3 pts) Repeat (i) for $V = 1$ V.



$\leftarrow V = \int_O^P \bar{E} \cdot d\bar{l} = 1\ V$

c) (7 pts) Determine the electrostatic potential function $V(x, y, z)$ corresponding to $\mathbf{E} = y\hat{x} + x\hat{y} + \cos(z)\hat{z}$ if $V(0, 0, 0) = 3$ V.

$-dV = \bar{E} \cdot d\bar{l} = y\,dx + x\,dy + \cos(z)\,dz = d(xy + \sin(z))$

$\hookrightarrow \therefore V(x, y, z) = -xy - \sin(z) + C$

$C = 3$ if $V(0, 0, 0) = 3$ V

$\hookrightarrow V = 3 - xy - \sin(z)$ V

3. Two parts of the following question are independent:

a) (15 pts) Consider the following *spherically* symmetric configuration of composite materials in steady-state equilibrium:

i. The region defined by $r \leq 1$ m, where $r = \sqrt{x^2 + y^2 + z^2}$ is the radial distance from the center of the spherical configuration, has $\epsilon = \epsilon_o$, $\sigma = 10^6$ S/m, and holds a net charge of $Q = +4$ C distributed uniformly over its spherical surface.

ii. A perfect dielectric shell with $\epsilon = 2\epsilon_o$ occupies the region $1 < r < 2$ m.

iii. Region $2 \leq r \leq 3$ m has the same material properties as region $r \leq 1$ m and holds zero net charge.

iv. Region $r > 3$ m is occupied by free space. In all four regions we have $\mu = \mu_o$.



Determine, in all four regions, (a) $\mathbf{D}$, (b) $\mathbf{E}$ and (c) $\mathbf{P}$, *and* (d) the surface charge densities, in C/m^2 units, at each of the three material boundaries at $r = 1$, 2, and 3 m. **Hint:** Make use of Gauss's law in integral form, $\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_V \rho dV$, with $\mathbf{D} = \epsilon\mathbf{E} = \epsilon_o\mathbf{E} + \mathbf{P}$, and a crucial fact about steady-state fields within conducting materials.

a) $\bar{D} = 0$, $r < 1$ m

$\bar{D} = \frac{4}{4\pi r^2}\hat{r} = \frac{\hat{r}}{\pi r^2}\ \frac{C}{m^2}$, $1 < r < 2$ m

$\bar{D} = \bar{0}$, $2 < r < 3$ m

$\bar{D} = \frac{\hat{r}}{\pi r^2}\ \frac{C}{m^2}$, $r > 3$ m

b) $\bar{E} = \bar{D}/\epsilon$:

$\bar{E} = 0$

$\bar{E} = \frac{\hat{r}}{2\pi\epsilon_0 r^2}\ \frac{V}{m}$

$\bar{E} = 0$

$\bar{E} = \frac{\hat{r}}{\pi\epsilon_0 r^2}\ \frac{C}{m^2}$

c) $\bar{P} = \bar{D} - \epsilon_0\bar{E}$

$\bar{P} = 0$

$\bar{P} = \frac{\hat{r}}{\pi\epsilon_0 r^2}$

$\bar{P} = 0$

$\bar{P} = 0$

b) (10 pts) Conducting plates placed on $z = 0$ and $z = 2$ m surfaces (of infinite extent) are separated by vacuum and maintained at potentials of $V(0) = 0$ and $V(2) = 4$ V. Also, the potential of the $z = 1$ m surface that supports some surface charge density ρ_{s1} is $V(1) = -2$ V.

i. What is the surface charge densities ρ_{s0} and ρ_{s2} at $z = 0$ and $z = 2$ m, respectively? Justify your answers.

ii. What is the surface charge density ρ_{s1} at $z = 1$ m? Justify your answer.

